NEET UG STANDARD — BIOLOGY TUTORIAL

# Plant Kingdom

Comprehensive Study Guide: Classification Systems, Algae, Bryophytes, Pteridophytes, Gymnosperms, and Angiosperms

## 1. Learning Objectives & Chapter Overview

Under the Five Kingdom classification proposed by R.H. Whittaker (1969), living organisms are grouped into **Monera, Protista, Fungi, Animalia, and Plantae**. This tutorial explores the detailed classification within **Kingdom Plantae** (popularly known as the plant kingdom).

**Core NEET UG Learning Objectives:**

- Understand the historical evolution of taxonomic systems: Artificial, Natural, and Phylogenetic classifications, along with Numerical Taxonomy, Cytotaxonomy, and Chemotaxonomy.
- Master the diagnostic features, pigment profiles, cell wall compositions, reserve food materials, and reproductive modes of the three classes of **Algae** (Chlorophyceae, Phaeophyceae, Rhodophyceae).
- Analyze the life cycle, gametophytic dominance, and ecological significance of **Bryophytes** (Liverworts and Mosses).
- Trace the origin of vascular tissues, sporophytic dominance, and the precursor to the seed habit (heterospory) in **Pteridophytes**.
- Examine the morphological adaptations, heterospory, dependent gametophytes, and naked-seed habit of **Gymnosperms**, and contrast them with **Angiosperms**.

## 2. Evolution of Classification Systems in Kingdom Plantae

Our scientific understanding of the plant kingdom has changed over time. Earlier classification systems placed all organisms possessing a cell wall into Kingdom Plantae. Today, **Fungi** and members of **Monera** and **Protista** having cell walls have been excluded from Plantae. Consequently, *cyanobacteria* (historically referred to as "blue-green algae") are no longer considered algae. Kingdom Plantae now strictly comprises **Algae, Bryophytes, Pteridophytes, Gymnosperms, and Angiosperms**.

### 2.1 Types of Classification Systems

- **Artificial Classification Systems:**
  - represent the earliest systems of classification.
  - Used only gross, superficial morphological characters such as habit, colour, number, and shape of leaves.
  - Based mainly on vegetative characters or on the **androecium structure** (such as the system given by **Linnaeus**).
  - **Limitations:** Separated closely related species because they were based on very few characteristics. Furthermore, they gave *equal weightage to vegetative and sexual characteristics*; this is unacceptable because vegetative characters are frequently and easily affected by the environment.

- **Natural Classification Systems:**
  - Developed based on natural affinities among organisms.
  - Considered not only external morphological features, but also internal features such as **ultrastructure, anatomy, embryology, and phytochemistry**.
  - A prominent natural classification for flowering plants was given by **George Bentham and Joseph Dalton Hooker**.

- **Phylogenetic Classification Systems:**
  - Currently acceptable systems based on **evolutionary relationships** between various organisms.
  - Assumes that organisms belonging to the same taxa share a common ancestor.

### 2.2 Modern Taxonomic Aids & Approaches

When supporting fossil evidence is absent, taxonomists use information from modern analytical sources to resolve difficulties in classification:

1. **Numerical Taxonomy:** Easily carried out using computers and based on *all observable characteristics*. Numbers and codes are assigned to all characters and the data are processed. In this way, **each character is given equal importance** and hundreds of characters can be considered simultaneously.
2. **Cytotaxonomy:** Based on cytological information such as **chromosome number, structure, and behaviour**.
3. **Chemotaxonomy:** Uses the **chemical constituents of the plant** to resolve taxonomic confusions.

## 3. Algae

Algae are chlorophyll-bearing, simple, thalloid, autotrophic, and largely aquatic (both freshwater and marine) organisms. They also occur in a variety of other habitats including moist stones, soils, and wood. Some occur in symbiotic associations:

- With fungi: **Lichens**
- With animals: e.g., on **sloth bear**

### 3.1 Form, Size, and Reproduction in Algae

The form and size of algae are highly variable:

- **Unicellular forms:** e.g., *Chlamydomonas* (motile), *Chlorella*.
- **Colonial forms:** e.g., *Volvox*.
- **Filamentous forms:** e.g., *Ulothrix* and *Spirogyra* (unbranched filaments), *Ectocarpus* (simple branched filamentous).
- **Massive marine plant bodies:** e.g., **Kelps** (profusely branched brown algae that may reach a height of $100\,\mathrm{m}$).

Algae reproduce by three main methods:

1. **Vegetative Reproduction:** Takes place by **fragmentation**. Each fragment develops into a thallus.
2. **Asexual Reproduction:** Occurs by the production of different types of spores, the most common being **zoospores**. Zoospores are flagellated (motile) and on germination give rise to new plants.
3. **Sexual Reproduction:** Takes place through the fusion of two gametes:
   - **Isogamous:** Fusion of two gametes that are similar in size. They may be:
     - Flagellated and similar in size: e.g., *Ulothrix*.
     - Non-flagellated (non-motile) but similar in size: e.g., *Spirogyra*.
   - **Anisogamous:** Fusion of two gametes dissimilar in size: e.g., species of *Eudorina*.
   - **Oogamous:** Fusion between one large, non-motile (static) female gamete and a smaller, motile male gamete: e.g., *Volvox*, *Fucus*. (Note: In red algae, sexual reproduction is oogamous, but both male and female gametes are non-motile).

### 3.2 Economic Importance of Algae

> - **Carbon Fixation & Oxygen Production:** At least **half of the total carbon dioxide ($\mathrm{CO_2}$) fixation on Earth** is carried out by algae through photosynthesis. Being photosynthetic, they increase the level of dissolved oxygen in their immediate aquatic environment.
> - **Primary Producers:** They are of paramount importance as primary producers of energy-rich compounds, forming the basis of the food cycles of all aquatic animals.

- **Marine Algae as Food:** Many species of *Porphyra*, *Laminaria*, and *Sargassum* are among the **70 species** of marine algae used as food.
- **Hydrocolloids:** Certain marine brown and red algae produce large amounts of hydrocolloids (water-holding substances) used commercially:
  - **Algin** — obtained from **brown algae**.
  - **Carrageen** — obtained from **red algae**.
- **Agar:** A commercial product obtained from red algae *Gelidium* and *Gracilaria*. Used to grow microbes and in the preparation of ice-creams and jellies.
- **Single-Cell Protein Supplement:** *Chlorella*, a unicellular alga rich in proteins, is used as a food supplement even by space travellers.

### 3.3 Classification of Algae

Algae are divided into three main classes based primarily on the type of pigment possessed and the type of stored food: **Chlorophyceae**, **Phaeophyceae**, and **Rhodophyceae**.

#### 3.3.1 Class I: Chlorophyceae (Green Algae)

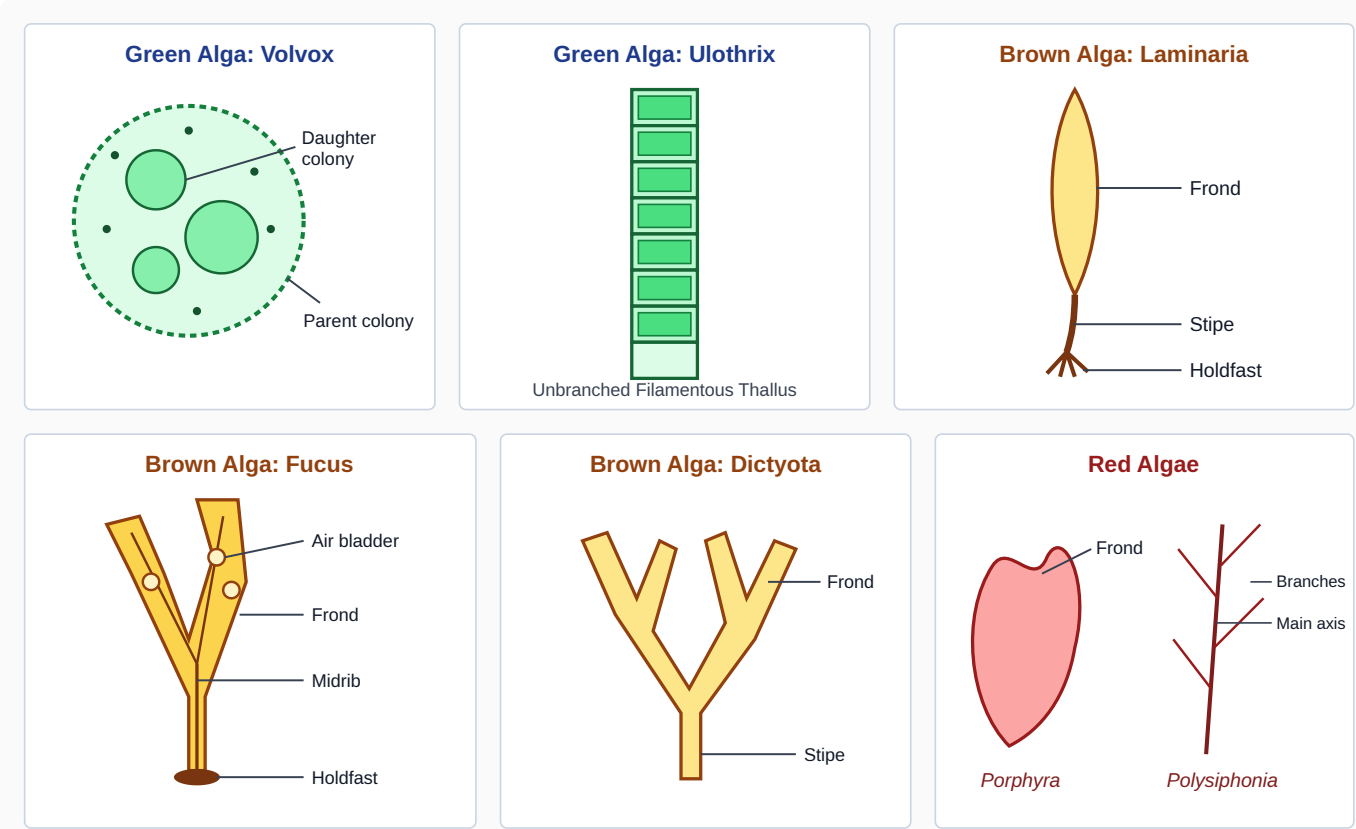
- **Plant Body:** Unicellular, colonial, or filamentous.
- **Pigments & Chloroplasts:** Grass-green due to the dominance of pigments **chlorophyll *a* and *b***. Pigments are localised in definite chloroplasts, which may be discoid, plate-like, reticulate, cup-shaped, spiral, or ribbon-shaped in different species.
- **Storage Bodies:** Most members have one or more storage bodies called **pyrenoids** located in the chloroplasts. *Pyrenoids contain protein besides starch.* Some algae may store food in the form of oil droplets.
- **Cell Wall:** Rigid cell wall made of an **inner layer of cellulose** and an **outer layer of pectose**.
- **Reproduction:** Vegetative by fragmentation; asexual by flagellated zoospores produced in zoosporangia; sexual reproduction shows considerable variation (isogamous, anisogamous, or oogamous).
- **Common Examples:** *Chlamydomonas*, *Volvox*, *Ulothrix*, *Spirogyra*, and *Chara*.

#### 3.3.2 Class II: Phaeophyceae (Brown Algae)

- **Habitat & Form:** Found primarily in **marine habitats**. Show great variation in size and form, ranging from simple branched, filamentous forms (*Ectocarpus*) to profusely branched forms represented by **kelps**, which may reach a height of $100\ \mathrm{m}$.
- **Pigments:** Possess **chlorophyll *a*, *c*, carotenoids, and xanthophylls**. They vary in colour from olive green to various shades of brown depending upon the amount of the xanthophyll pigment **fucoxanthin** present in them.
- **Stored Food:** Complex carbohydrates in the form of **laminarin or mannitol**.
- **Cell Structure & Wall:** Vegetative cells have a **cellulosic wall** usually covered on the outside by a gelatinous coating of **algin**. The protoplast contains plastids, a centrally located vacuole, and a nucleus.
- **Thallus Differentiation:** The plant body is usually attached to the substratum by a **holdfast**, has a stalk called the **stipe**, and a leaf-like photosynthetic organ called the **frond**.
- **Reproduction:** Vegetative by fragmentation. Asexual reproduction in most brown algae is by **biflagellate zoospores** that are pear-shaped (pyriform) and have **two unequal laterally attached flagella**. Sexual reproduction may be isogamous, anisogamous, or oogamous. Union of gametes may take place in water or within the oogonium (in oogamous species). The gametes are also **pyriform (pear-shaped)** and bear two laterally attached flagella.
- **Common Examples:** *Ectocarpus*, *Dictyota*, *Laminaria*, *Sargassum*, and *Fucus*.

### 3.3.3 Class III: Rhodophyceae (Red Algae)

- **Pigments:** Commonly called red algae because of the predominance of the red pigment, **r-phycoerythrin**, along with **chlorophyll *a* and *d***.
- **Habitat:** Majority are **marine** with greater concentrations found in warmer areas. They occur in both well-lighted regions close to the surface of water and at **great depths in oceans** where relatively little light penetrates.
- **Plant Body & Stored Food:** The red thalli of most red algae are multicellular, and some have complex body organisation. Food is stored as **floridean starch**, which is *very similar to amylopectin and glycogen in structure*.
- **Cell Wall:** Made of cellulose, pectin, and **polysulphate esters**.
- **Reproduction:** Vegetatively by fragmentation. Asexually by **non-motile spores** and sexually by **non-motile gametes** (flagella are completely absent). Sexual reproduction is **oogamous** and accompanied by **complex post-fertilisation developments**.
- **Common Examples:** *Polysiphonia*, *Porphyra*, *Gracilaria*, and *Gelidium*.




**Tutorial Diagram 1: Representative Members of Green Algae (Volvox, Ulothrix), Brown Algae (Laminaria, Fucus, Dictyota), and Red Algae (Porphyra, Polysiphonia) showing diagnostic morphological structures.**

### 3.4 Comparative Summary of Algal Divisions

| Classes | Common Name | Major Pigments | Stored Food | Cell Wall | Flagellar Number & Position of Insertions | Habitat |
|---|---|---|---|---|---|---|
| **Chlorophyceae** | Green algae | Chlorophyll *a, b* | Starch (pyrenoids contain protein besides starch; some store oil droplets) | Cellulose (inner) and pectose (outer) | 2–8, equal, apical | Fresh water, brackish water, salt water |
| **Phaeophyceae** | Brown algae | Chlorophyll *a, c*, fucoxanthin (xanthophyll), carotenoids | Mannitol, laminarin | Cellulose and algin | 2, unequal, lateral | Fresh water (rare), brackish water, salt water |
| **Rhodophyceae** | Red algae | Chlorophyll *a, d*, r-phycoerythrin | Floridean starch | Cellulose, pectin and polysulphate esters | Absent | Fresh water (some), brackish water, salt water (most) |

## 4. Bryophytes

Bryophytes include the various **mosses and liverworts** that are found commonly growing in moist, shaded areas in the hills. Bryophytes are called the **"amphibians of the plant kingdom"** because these plants can live in soil but are strictly dependent on water for sexual reproduction.

### 4.1 General Characteristics & Life Cycle

- **Habitat & Ecology:** Occur in damp, humid, and shaded localities. They play an important role in **plant succession on bare rocks/soil**.
- **Plant Body Differentiation:** More differentiated than that of algae. It is thallus-like and prostrate or erect, and attached to the substratum by **unicellular or multicellular rhizoids**. They **lack true roots, stem, or leaves**, though they may possess root-like, leaf-like, or stem-like structures.
- **Dominant Phase (Gametophyte):** The main plant body of the bryophyte is **haploid ($n$)**. Because it produces gametes, it is called a **gametophyte**.
- **Sex Organs:** Multicellular.
  - **Antheridium (Male sex organ):** Produces **biflagellate antherozoids** (motile male gametes).
  - **Archegonium (Female sex organ):** Flask-shaped structure that produces a **single egg**.
- **Fertilisation & Sporophyte Development:** Antherozoids are released into water where they come in contact with the archegonium. An antherozoid fuses with the egg to produce the diploid **zygote ($2n$)**. *Zygotes do not undergo reduction division (meiosis) immediately.* Instead, they undergo mitotic divisions to produce a multicellular diploid body called a **sporophyte ($2n$)**.
- **Dependency of Sporophyte:** The sporophyte is **not free-living**; it remains attached to the photosynthetic gametophyte and derives nourishment from it. It is differentiated into a **foot, seta, and capsule**.
- **Spore Formation:** Some cells of the sporophyte (spore mother cells inside the capsule) undergo **reduction division (meiosis)** to produce haploid spores ($n$). These spores germinate to produce the haploid gametophyte.

### 4.2 Economic & Ecological Importance of Bryophytes

- **Food Source:** Some mosses provide food for herbaceous mammals, birds, and other animals.
- **Peat & Trans-shipment:** Species of *Sphagnum* (a moss) provide **peat** that has long been used as fuel, and as **packing material for trans-shipment of living material** because of its high capacity to hold water.

- **Ecological Pioneers:** Mosses along with lichens are the **first organisms to colonise rocks**. They decompose rocks, making the substrate suitable for the growth of higher plants.
- **Soil Conservation:** Because mosses form dense mats on the soil, they reduce the impact of falling rain and **prevent soil erosion**.

### 4.3 Classification of Bryophytes: Liverworts vs. Mosses

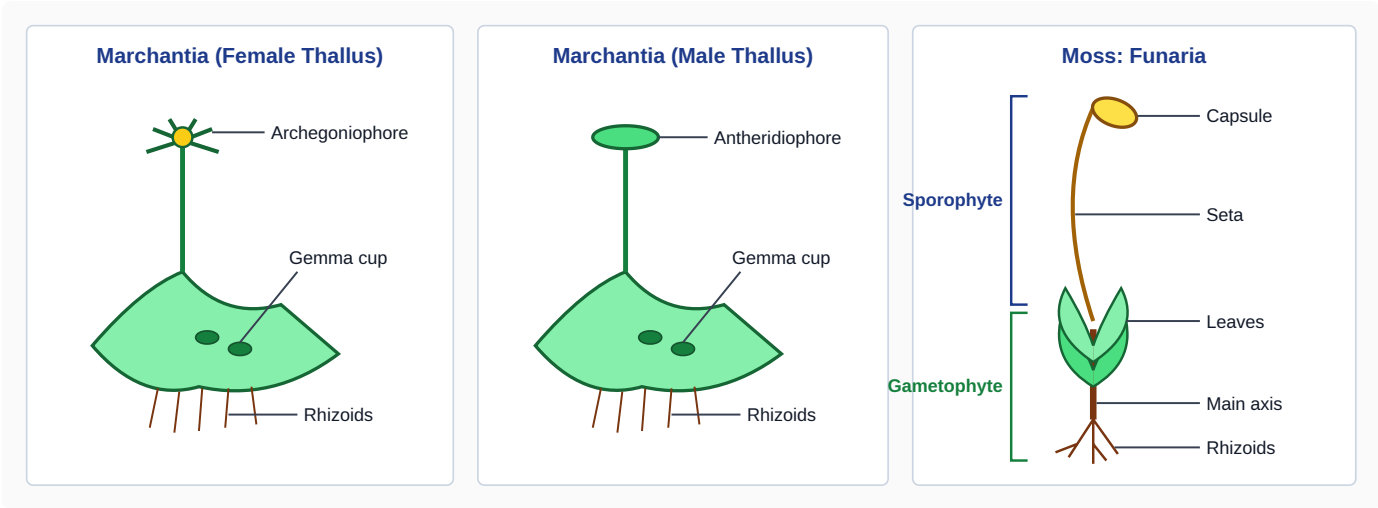
Bryophytes are divided into two groups: **Liverworts** and **Mosses**.

#### 4.3.1 Liverworts

- **Habitat:** Moist, shady habitats such as banks of streams, marshy ground, damp soil, bark of trees, and deep in the woods.
- **Plant Body:** Thalloid, e.g., *Marchantia*. The thallus is **dorsiventral** and closely appressed to the substrate. Leafy members have tiny leaf-like appendages in **two rows** on the stem-like structures. Rhizoids are unicellular.
- **Asexual Reproduction:** By fragmentation of thalli, or by the formation of specialised structures called **gemmae** (singular: *gemma*).
  - **Gemmae** are **green, multicellular, asexual buds**, which develop in small receptacles called **gemma cups** located on the thalli.
  - Gemmae detach from the parent body and germinate to form new individuals.
- **Sexual Reproduction:** Male and female sex organs are produced either on the **same or on different thalli** (in *Marchantia*, male and female thalli are separate, bearing antheridiophores and archegoniophores, respectively). The sporophyte is differentiated into a foot, seta, and capsule. After meiosis, spores are produced within the capsule, which germinate to form free-living gametophytes.

#### 4.3.2 Mosses

- **Two-Stage Gametophyte:** The predominant stage of the life cycle of a moss is the gametophyte, which consists of two distinct stages:
  1. **Protonema stage:** The first stage, which develops **directly from a spore**. It is a creeping, green, branched, and frequently filamentous stage.
  2. **Leafy stage:** The second stage, which develops from the **secondary protonema as a lateral bud**. It consists of upright, slender axes bearing **spirally arranged leaves**. It is attached to the soil through **multicellular and branched rhizoids**. *This stage bears the sex organs.*
- **Reproduction:**
  - **Vegetative reproduction:** By fragmentation and **budding in the secondary protonema**.
  - **Sexual reproduction:** Antheridia and archegonia are produced at the **apex of the leafy shoots**. After fertilisation, the zygote develops into a sporophyte consisting of a foot, seta, and capsule.
- **Elaborate Sporophyte:** *The sporophyte in mosses is more elaborate than that in liverworts.* The capsule contains spores formed after meiosis, and mosses possess an **elaborate mechanism of spore dispersal**.
- **Common Examples:** *Funaria*, *Polytrichum*, and *Sphagnum*.




**Tutorial Diagram 2: Bryophytes — Female and Male Thalli of Liverwort (Marchantia) with Gemma Cups, and Funaria (Moss) illustrating the attached Sporophyte (Foot, Seta, Capsule) on the leafy Gametophyte.**

## 5. Pteridophytes

Pteridophytes include **horsetails and ferns**. Evolutionarily, they are the **first terrestrial plants to possess vascular tissues — xylem and phloem**. They are used for medicinal purposes, as **soil-binders**, and are frequently grown as ornamentals.

### 5.1 Sporophytic Dominance & Plant Body

- **Habitat:** Found in cool, damp, shady places, though some may flourish well in sandy-soil conditions.
- **Dominant Phase (Sporophyte):** Unlike bryophytes (where the dominant phase is the gametophyte), in pteridophytes the **main plant body is a diploid sporophyte ($2n$)** which is differentiated into **true root, stem, and leaves**. These organs possess well-differentiated vascular tissues.
- **Leaf Types:**
  - **Microphylls (small leaves):** e.g., *Selaginella*.
  - **Macrophylls (large leaves):** e.g., Ferns.
- **Sporangia, Sporophylls, and Strobili:**
  - Sporophytes bear **sporangia** that are subtended by leaf-like appendages called **sporophylls**.
  - In some cases, sporophylls may form distinct compact structures called **strobili or cones** (e.g., *Selaginella*, *Equisetum*).

### 5.2 Gametophyte (Prothallus) & Geographical Restriction

- The sporangia produce haploid spores ($n$) by **meiosis in spore mother cells ($2n$)**.
- The spores germinate to give rise to inconspicuous, small, but multicellular, free-living, mostly photosynthetic thalloid gametophytes called **prothallus**.
- **Why is the spread of living pteridophytes limited and restricted to narrow geographical regions?**
  1. The gametophytes (prothallus) require specific, restricted **cool, damp, shady places to grow**.
  2. There is a strict **need for water for fertilisation** (for the transfer of motile male gametes, the antherozoids, released from the antheridia to the mouth of the archegonium).
- Fusion of the male gamete with the egg present in the archegonium results in the formation of a diploid **zygote ($2n$)**, which develops into a multicellular, well-differentiated sporophyte.

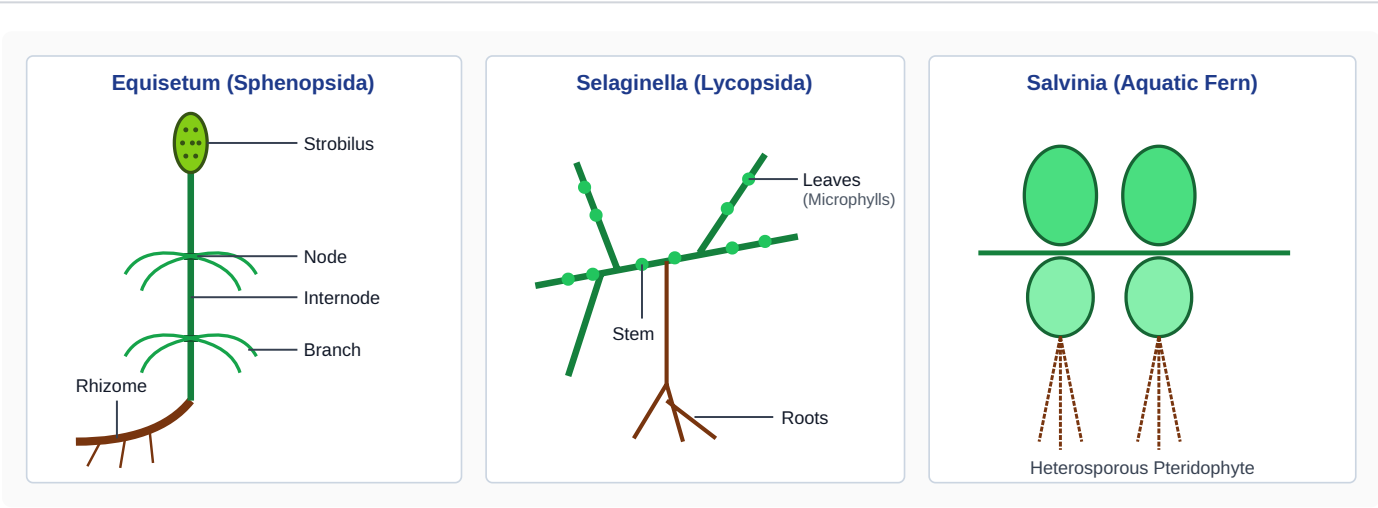
### 5.3 Homospory vs. Heterospory & Precursor to Seed Habit

- **Homosporous Pteridophytes:** In the **majority of pteridophytes**, all spores produced are of similar kinds (e.g., *Lycopodium*, *Equisetum*, *Dryopteris*, *Pteris*, *Adiantum*, *Psilotum*).

- **Heterosporous Pteridophytes:** Genera like ***Selaginella*** **and** ***Salvinia*** produce two kinds of spores: **macro (large) / megaspores** and **micro (small) spores**.
- **Evolutionary Significance (Precursor to Seed Habit):**
  1. Megaspores and microspores germinate and give rise to **female and male gametophytes**, respectively.
  2. The **female gametophytes in these plants are retained on the parent sporophytes** for variable periods.
  3. The **development of the zygotes into young embryos takes place within the female gametophytes**.
  4. This event is a **precursor to the seed habit**, considered an important step in plant evolution.

### 5.4 Four Classes of Pteridophytes

| Class | Representative Genera (Examples) |
|---|---|
| **1. Psilopsida** | *Psilotum* |
| **2. Lycopsida** | *Selaginella*, *Lycopodium* |
| **3. Sphenopsida** | *Equisetum* (horsetail) |
| **4. Pteropsida** | *Dryopteris*, *Pteris*, *Adiantum* (ferns) |




**Tutorial Diagram 3: Representative Pteridophytes — Equisetum (showing terminal strobilus, nodes, internodes, and rhizome), Selaginella (showing microphylls), and Salvinia (aquatic heterosporous pteridophyte).**

## 6. Gymnosperms

The gymnosperms (*gymnos*: naked; *sperma*: seeds) are plants in which the **ovules are not enclosed by any ovary wall** and remain exposed, both before and after fertilisation. Consequently, the seeds that develop post-fertilisation are not covered (i.e., they are **naked**).

### 6.1 Habit, Root, Stem, and Leaf Adaptations

- **Habit & Size:** Include medium-sized trees, tall trees, and shrubs. One of the gymnosperms, the giant redwood tree ***Sequoia***, is one of the tallest tree species.
- **Root System:** Generally **tap roots**. Specialised root associations occur in specific genera:
  - **Mycorrhiza (Fungal association):** Present in ***Pinus***.
  - **Coralloid Roots:** Small specialised roots in ***Cycas*** associated with $N_2$**-fixing cyanobacteria**.
- **Stem Branching:**
  - **Unbranched stem:** *Cycas*.

- **Branched stem:** *Pinus*, *Cedrus* (and *Ginkgo*, which bears long shoots and dwarf shoots).

- **Leaves & Xerophytic Adaptations:**
  - Leaves may be simple or compound. In *Cycas*, the **pinnate leaves** persist for a few years.
  - Gymnosperm leaves are well-adapted to withstand extremes of temperature, humidity, and wind.
  - In **conifers** (e.g., *Pinus*), the **needle-like leaves reduce the surface area**. Their **thick cuticle** and **sunken stomata** also help to reduce water loss.

### 6.2 Heterospory, Strobili, and Reproduction

- All gymnosperms are **heterosporous**; they produce haploid **microspores** and **megaspores** within sporangia borne on sporophylls, which are arranged spirally along an axis to form lax or compact **strobili or cones**.
- **Male Strobili (Microsporangiate Cones):**
  - Bear **microsporophylls** and **microsporangia**.
  - Microspores develop into a male gametophytic generation which is **highly reduced and confined to only a limited number of cells**. This reduced male gametophyte is called a **pollen grain**.
  - Development of pollen grains takes place within the microsporangia.

- **Female Strobili (Macrosporangiate Cones):**
  - Bear **megasporophylls** with **ovules (megasporangia)**.
  - **Monoecious vs. Dioecious Condition:**
    - In ***Pinus***, male and female cones (strobili) are borne on the **same tree** (monoecious).
    - In ***Cycas***, male cones and megasporophylls are borne on **different trees** (dioecious). Note that in *Cycas*, megasporophylls do not form a compact female cone.

- **Ovule & Female Gametophyte Development:**
  - The **megaspore mother cell ($2n$)** is differentiated from one of the cells of the **nucellus**.
  - The nucellus is protected by envelopes, and this composite structure is called an **ovule**.
  - The megaspore mother cell divides **meiotically** to form **four haploid megaspores ($n$)**.
  - One of the megaspores enclosed within the megasporangium (nucellus) develops into a **multicellular female gametophyte ($n$)** that bears **two or more archegonia** (female sex organs). The multicellular female gametophyte is retained within the megasporangium.

- **Non-Free-Living Gametophytes, Pollination & Fertilisation:**
  - **Crucial Distinction:** *Unlike bryophytes and pteridophytes, in gymnosperms the male and female gametophytes do not have an independent free-living existence.* They remain within the sporangia retained on the sporophytes.
  - Pollen grains are released from the microsporangium and carried in **air currents (anemophily / wind pollination)**, coming in contact with the opening of the ovules borne on megasporophylls.
  - A **pollen tube** carrying the male gametes grows towards the archegonia in the ovules and discharges its contents near the mouth of the archegonia (siphonogamy — water is not required for fertilisation).
  - Following fertilisation, the **zygote develops into an embryo** and the **ovules into uncovered (naked) seeds**.

## 7. Angiosperms

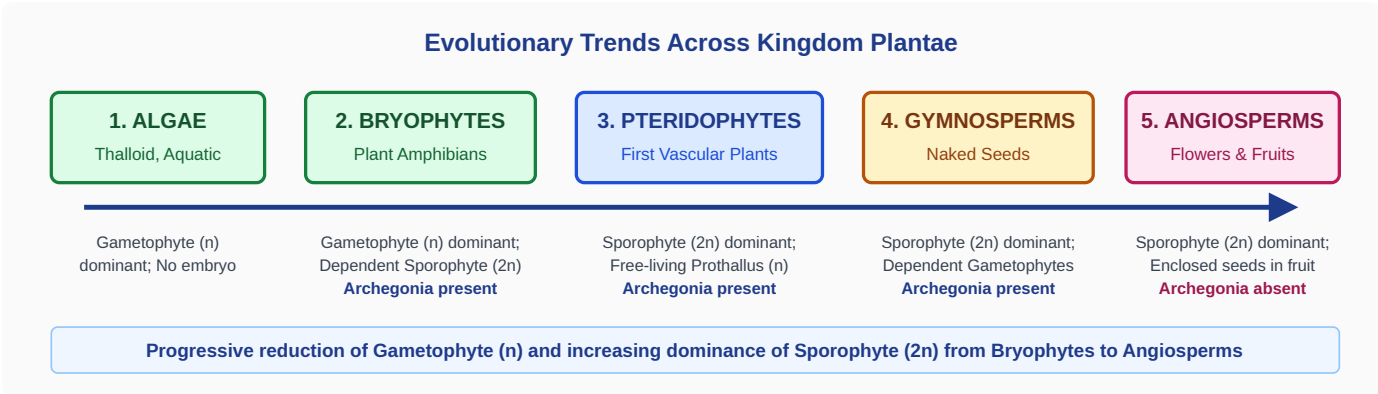
Unlike gymnosperms where the ovules are naked, in the **angiosperms (flowering plants)** the pollen grains and ovules are developed in specialised structures called **flowers**, and the **seeds are enclosed in fruits**.

- **Diversity & Size Range:** Angiosperms are an exceptionally large group of plants occurring in a wide range of habitats. They range in size from the smallest microscopic aquatic plant ***Wolffia*** to tall trees of ***Eucalyptus*** (over $100\,\mathrm{m}$).
- **Importance:** Provide food, fodder, fuel, medicines, and several other commercially important products.
- **Classification:** Divided into two classes:
  1. **Dicotyledons**

2. **Monocotyledons**

## 8. Conceptual Connections & Comparative Synthesis across Plantae

A frequent area of assessment in NEET UG involves comparing the site of reduction division (meiosis), the ploidy of specific plant structures, the presence/absence of archegonia, and evolutionary trends across the five plant groups.




**Tutorial Diagram 4: Evolutionary Progression across Kingdom Plantae showing Gametophyte-Sporophyte relationships and Archegonial distribution.**

### 8.1 Site and Timing of Reduction Division (Meiosis)

| Plant Group | When & Where Reduction Division (Meiosis) Takes Place |
|---|---|
| **Liverwort (Bryophyte)** | During spore formation, in the **spore mother cells inside the capsule** of the sporophyte. |
| **Moss (Bryophyte)** | During spore formation, in the **spore mother cells inside the capsule** of the sporophyte. |
| **Fern (Pteridophyte)** | During spore formation, in the **spore mother cells inside the sporangia** borne on sporophylls of the sporophyte. |
| **Gymnosperm** | During microspore and megaspore formation: in the **microspore mother cells (in microsporangia)** and the **megaspore mother cell (in the nucellus/megasporangium)**. |
| **Angiosperm** | During microspore (pollen) and megaspore formation: in the **microspore mother cells (in anther)** and the **megaspore mother cell (in ovule)**. |

### 8.2 Master Ploidy Reference Table

| Plant Structure / Cell | Ploidy Level | Reason / Developmental Origin |
|---|---|---|
| Protonemal cell of a moss | Haploid ($n$) | Develops directly from a haploid spore ($n$) by mitosis; part of the gametophyte. |
| Leaf cell of a moss | Haploid ($n$) | Belongs to the leafy gametophyte stage of a moss. |
| Gemma cell in *Marchantia* | Haploid ($n$) | Multicellular asexual bud produced mitotically on the haploid gametophytic thallus. |
| Ovum of a liverwort | Haploid ($n$) | Female gamete produced inside the archegonium of the haploid gametophyte. |
| Prothallus cell of a fern | Haploid ($n$) | Free-living thalloid gametophyte formed by the germination of a haploid spore. |
| Zygote of a fern | Diploid ($2n$) | Formed by the fusion (syngamy) of a haploid antherozoid ($n$) and a haploid egg ($n$). |
| Meristem cell of a monocot | Diploid ($2n$) | Actively dividing vegetative cell of the dominant angiosperm sporophyte ($2n$). |
| Primary Endosperm Nucleus (PEN) in a dicot | Triploid ($3n$) | Formed by triple fusion (fusion of one haploid male gamete $n$ with two haploid polar nuclei $n + n$) in angiosperms. |
| Female gametophyte (endosperm) in gymnosperms | Haploid ($n$) | Develops from a functional haploid megaspore ($n$) *before* fertilisation. |

## 9. Worked Examples (Concept Application)

#### Worked Example 1: Ploidy & Chromosome Number Calculation

**Problem:** In the moss *Funaria*, if a leaf cell contains $10$ chromosomes, determine the number of chromosomes present in: (a) a protonemal cell, (b) an antherozoid, (c) a spore mother cell in the capsule, and (d) a spore.

**Solution:**

- In mosses (bryophytes), the leafy stage is part of the dominant **haploid gametophyte ($n$)**. Therefore, the chromosome number of the leaf cell gives $n = 10$.
- **(a) Protonemal cell:** The protonema is the first stage of the haploid gametophyte ($n$). Thus, chromosome number $= n = 10$.
- **(b) Antherozoid:** The male gamete is produced by mitosis in the haploid antheridium ($n$). Thus, chromosome number $= n = 10$.
- **(c) Spore mother cell in the capsule:** The capsule is part of the diploid sporophyte ($2n$), so spore mother cells are diploid ($2n$). Thus, chromosome number $= 2n = 2 \times 10 = 20$.
- **(d) Spore:** Spores are formed after reduction division (meiosis) of spore mother cells, making them haploid ($n$). Thus, chromosome number $= n = 10$.

### Worked Example 2: Comparing Red Algae vs. Brown Algae and Liverworts vs. Mosses

**Problem:** State three clear diagnostic differences between (i) Red algae and Brown algae, and (ii) Liverworts and Mosses.

**Solution:**

**(i) Red Algae (Rhodophyceae) vs. Brown Algae (Phaeophyceae):**

1. **Major Pigments:** Red algae possess *chlorophyll a, d* and *r-phycoerythrin*; Brown algae possess *chlorophyll a, c*, carotenoids, and *fucoxanthin*.
2. **Stored Food:** Red algae store food as *floridean starch* (structurally similar to amylopectin and glycogen); Brown algae store food as complex carbohydrates *laminarin or mannitol*.
3. **Motile Stages / Flagella:** Red algae completely lack flagella (spores and gametes are non-motile); Brown algae produce pear-shaped (pyriform) zoospores and gametes bearing *two unequal, laterally attached flagella*.

**(ii) Liverworts vs. Mosses:**

1. **Gametophyte Structure:** Liverworts have a dorsiventral thalloid body (or leafy axis with leaves in two rows) and *unicellular rhizoids*; Mosses have a two-stage gametophyte (creeping filamentous *protonema* and upright *leafy stage* with spirally arranged leaves) and *multicellular, branched rhizoids*.
2. **Specialised Asexual Buds:** Liverworts (such as *Marchantia*) reproduce asexually by green, multicellular buds called *gemmae* borne in gemma cups; Mosses reproduce vegetatively by fragmentation and budding in the secondary protonema.
3. **Sporophyte Complexity:** The sporophyte in mosses is *more elaborate* than that in liverworts and possesses an elaborate mechanism of spore dispersal.

### Worked Example 3: Key Term Definitions with Examples

**Problem:** Define the following botanical terms with suitable examples: (a) Protonema, (b) Sporophyll, (c) Isogamy, and (d) Archegonium.

**Solution:**

- **(a) Protonema:** The creeping, green, branched, and frequently filamentous first stage of the gametophyte in mosses that develops directly from a haploid spore. *Example: Funaria*, *Polytrichum*, *Sphagnum*.
- **(b) Sporophyll:** Leaf-like appendages on the sporophyte of pteridophytes and gymnosperms that subtend or bear sporangia. In some species, they aggregate into compact strobili or cones. *Example: Selaginella*, *Equisetum*, *Pinus*, *Cycas*.
- **(c) Isogamy:** Sexual reproduction involving the fusion of two gametes that are morphologically similar in size. They may be flagellated (motile) as in *Ulothrix*, or non-flagellated (non-motile) as in *Spirogyra*.
- **(d) Archegonium:** The multicellular, flask-shaped female sex organ that produces a single egg cell. It is present in **Bryophytes, Pteridophytes, and Gymnosperms** (absent in Algae and Angiosperms).

## 10. NEET UG Practice Questions with Worked Solutions

*Note: All questions below are original practice questions designed strictly in accordance with the NEET UG syllabus and examination pattern.*

**Practice Question 1** Single-Choice MCQ

Why was the artificial system of plant classification proposed by Linnaeus considered unacceptable by later taxonomists?

(A) It relied exclusively on internal anatomical, embryological, and phytochemical features.

(B) It gave equal weightage to vegetative and sexual characteristics, even though vegetative characters are easily affected by the environment.

(C) It required complex computer processing of hundreds of observable characters simultaneously.

(D) It grouped organisms solely on the basis of chromosome number and meiotic behaviour.

**Correct Answer: (B)**

**Worked Solution:** Artificial classification systems (such as the one given by Linnaeus based on androecium structure or superficial morphological traits) separated closely related species because they relied on very few characteristics. Crucially, they gave equal weightage to vegetative and sexual characteristics, which is unacceptable because vegetative characters are much more easily affected by environmental conditions.

**Practice Question 2** Statement-Based MCQ

Read the following statements regarding sexual reproduction in algae and select the correct option:

**Statement I:** Both *Ulothrix* and *Spirogyra* exhibit isogamous sexual reproduction, but the gametes of *Ulothrix* are flagellated while those of *Spirogyra* are non-flagellated.
**Statement II:** Species of *Eudorina* exhibit oogamous reproduction, whereas *Volvox* and *Fucus* exhibit anisogamous reproduction.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

**Correct Answer: (C)**

**Worked Solution:** Fusion of gametes similar in size is called isogamy. In *Ulothrix*, the gametes are flagellated (motile) and similar in size; in *Spirogyra*, the gametes are non-flagellated (non-motile) and similar in size. Hence, Statement I is correct. Statement II is incorrect because species of *Eudorina* show **anisogamous** reproduction (fusion of two gametes dissimilar in size), whereas *Volvox* and *Fucus* show **oogamous** reproduction (fusion between one large non-motile female gamete and a smaller motile male gamete).

**Practice Question 3** Single-Choice MCQ

A marine biologist collects an alga from warm ocean waters that has a multicellular thallus, lacks flagellated spores or gametes throughout its life cycle, and stores food in a form structurally very similar to amylopectin and glycogen. Which of the following sets of pigments and cell wall components is characteristic of this alga?

(A) Chlorophyll *a*, *c*, fucoxanthin; cell wall of cellulose and algin

(B) Chlorophyll *a*, *d*, r-phycoerythrin; cell wall of cellulose, pectin, and polysulphate esters

(C) Chlorophyll *a*, *b*; cell wall of inner cellulose and outer pectose

(D) Chlorophyll *a*, *d*, fucoxanthin; cell wall of cellulose and carrageen only

**Correct Answer: (B)**

**Worked Solution:** An alga that lacks flagellated stages (non-motile spores and non-motile gametes) and stores food as **floridean starch** (which is very similar to amylopectin and glycogen in structure) belongs to the class **Rhodophyceae (red algae)**. Members of Rhodophyceae contain the major pigments **chlorophyll *a*, *d*, and r-phycoerythrin**, and their cell wall is composed of **cellulose, pectin, and polysulphate esters**.

**Practice Question 4** Matching MCQ

Match the organisms in Column I with their respective commercial/economic significance or characteristic structure in Column II:

| Column I | Column II |
|---|---|
| (a) *Gelidium* and *Gracilaria* | (i) Unicellular protein-rich alga used by space travellers |
| (b) *Chlorella* | (ii) Provides peat used as fuel and water-holding packing material |
| (c) *Sphagnum* | (iii) Source of agar used to grow microbes and prepare jellies |
| (d) *Ectocarpus* | (iv) Simple branched, filamentous marine brown alga |

(A) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(D) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

**Correct Answer: (A)**

**Worked Solution:**
• *Gelidium* and *Gracilaria* are red algae from which commercial **agar** is obtained, used to grow microbes and in preparations of ice-creams and jellies: (a)-(iii).
• *Chlorella* is a unicellular green alga rich in proteins, used as a food supplement by space travellers: (b)-(i).
• *Sphagnum* is a moss that provides **peat** used as fuel and as packing material for trans-shipment of living material: (c)-(ii).
• *Ectocarpus* is a simple branched, filamentous brown alga (Phaeophyceae): (d)-(iv).

**Practice Question 5** Multi-Statement Evaluation

Which of the following statements are **correct** regarding the life cycle and structure of mosses?

1. The protonema stage develops directly from a haploid spore and is a creeping, green, branched, and frequently filamentous stage.
2. The leafy stage develops from the primary protonema as an apical bud.
3. The leafy stage consists of upright, slender axes bearing spirally arranged leaves and is attached to the soil by unicellular, unbranched rhizoids.
4. Vegetative reproduction in mosses takes place by fragmentation and budding in the secondary protonema.
5. The sex organs (antheridia and archegonia) are produced at the apex of the leafy shoots.

(A) 1, 2, and 3 only

(B) 1, 4, and 5 only

(C) 2, 3, and 4 only

(D) 1, 3, 4, and 5 only

**Correct Answer: (B)**

**Worked Solution:**
• Statement 1 is **correct**: The first stage of the moss gametophyte is the protonema stage, which develops directly from a spore and is creeping, green, branched, and frequently filamentous.
• Statement 2 is **incorrect**: The leafy stage develops from the *secondary protonema* as a *lateral bud* (not primary protonema / apical bud).
• Statement 3 is **incorrect**: The leafy stage of mosses is attached to the soil through *multicellular and branched rhizoids* (unicellular rhizoids occur in liverworts).
• Statements 4 and 5 are **correct**: Mosses reproduce vegetatively by fragmentation and budding in the secondary protonema, and sex organs are produced at the apex of the leafy shoots.

**Practice Question 6** Assertion–Reason

**Assertion (A):** The spread of living pteridophytes is limited and restricted to narrow geographical regions.
**Reason (R):** Their gametophytes (prothallus) require cool, damp, shady places to grow, and water is required for the transfer of antherozoids to the mouth of the archegonium for fertilisation.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

**Correct Answer: (A)**

**Worked Solution:** Even though the sporophyte of pteridophytes possesses vascular tissues (xylem and phloem) and true roots, stems, and leaves, their free-living thalloid gametophyte (prothallus) requires cool, damp, shady habitats to grow, and water is essential for the swimming of motile antherozoids to the archegonium during fertilisation. Because of these two specific requirements, the geographical distribution of living pteridophytes is restricted.

**Practice Question 7** Assertion–Reason

**Assertion (A):** Heterospory in genera like *Selaginella* and *Salvinia* is considered a precursor to the seed habit, an important step in evolution.
**Reason (R):** In heterosporous pteridophytes, the female gametophytes are retained on the parent sporophytes for variable periods, and the development of the zygotes into young embryos takes place within the female gametophytes.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

**Correct Answer: (A)**

**Worked Solution:** Genera like *Selaginella* and *Salvinia* produce two kinds of spores — macro (large) and micro (small) spores — a condition called heterospory. The megaspores and microspores germinate into female and male gametophytes, respectively. Crucially, the female gametophytes are retained on the parent sporophytes for variable periods, and the development of the zygotes into young embryos occurs within the female gametophytes. This retention and in-situ embryo development is the precursor to the seed habit.

**Practice Question 8** Single-Choice MCQ

Examine the following comparison between the two gymnosperm genera *Pinus* and *Cycas*. Which row in the table below is completely accurate?

| Option | Feature in *Pinus* | Feature in *Cycas* |
|---|---|---|
| (A) | Unbranched stem; coralloid roots with cyanobacteria | Branched stem; mycorrhizal roots with fungi |
| (B) | Branched stem; mycorrhizal roots; male and female cones on the same tree | Unbranched stem; coralloid roots with $N_2$-fixing cyanobacteria; male cones and megasporophylls on different trees |
| (C) | Male and female cones borne on different trees (dioecious) | Male and female cones borne on the same tree (monoecious) |
| (D) | Free-living independent female gametophyte | Ovules enclosed within an ovary wall |

(A) Row (A)

(B) Row (B)

(C) Row (C)

(D) Row (D)

**Correct Answer: (B)**

**Worked Solution:** In *Pinus*, the stem is branched (also in *Cedrus*), roots have fungal association in the form of **mycorrhiza**, and male and female cones are borne on the **same tree**. In *Cycas*, the stem is **unbranched**, pinnate leaves persist for a few years, specialised **coralloid roots** are associated with $N_2$-fixing cyanobacteria, and male cones and megasporophylls are borne on **different trees**.

**Practice Question 9** Matching MCQ

Match the plant genera in Column I with their correct taxonomic group/class in Column II:

| Column I | Column II |
|---|---|
| (a) *Chlamydomonas* | (i) Moss (Bryophyte) |
| (b) *Cycas* | (ii) Lycopsida (Pteridophyte) |
| (c) *Selaginella* | (iii) Chlorophyceae (Algae) |
| (d) *Sphagnum* | (iv) Gymnosperm |
| (e) *Adiantum* | (v) Pteropsida (Pteridophyte) |

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i), (e)-(v)

(B) (a)-(iii), (b)-(iv), (c)-(v), (d)-(i), (e)-(ii)

(C) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii), (e)-(v)

(D) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i), (e)-(v)

**Correct Answer: (A)**

**Worked Solution:**
• *Chlamydomonas* is a unicellular green alga (Chlorophyceae): (a)-(iii).
• *Cycas* is a naked-seeded plant (Gymnosperm): (b)-(iv).
• *Selaginella* belongs to class Lycopsida of Pteridophytes: (c)-(ii).
• *Sphagnum* is a peat moss (Bryophyte): (d)-(i).
• *Adiantum* (along with *Dryopteris* and *Pteris*) belongs to class Pteropsida of Pteridophytes: (e)-(v).

**Practice Question 10** Single-Choice MCQ

In which of the following groups of plants do both the male and female gametophytes lack an independent free-living existence and remain within the sporangia retained on the sporophytes?

(A) Liverworts and Mosses

(B) Homosporous Pteridophytes such as *Equisetum* and *Dryopteris*

(C) Heterosporous Pteridophytes such as *Selaginella* and *Salvinia*

(D) Gymnosperms such as *Pinus* and *Cycas*

**Correct Answer: (D)**

**Worked Solution:** Unlike bryophytes and pteridophytes, in **gymnosperms** (and angiosperms), the male and the female gametophytes do not have an independent free-living existence. They remain within the sporangia (microsporangia and megasporangia) retained on the sporophytes.

**Practice Question 11** Single-Choice MCQ

Consider the following eight plant structures/cells:
(i) Protonemal cell of a moss
(ii) Primary endosperm nucleus (PEN) in a dicot
(iii) Leaf cell of a moss
(iv) Prothallus cell of a fern
(v) Gemma cell in *Marchantia*
(vi) Meristem cell of a monocot
(vii) Ovum of a liverwort
(viii) Zygote of a fern
How many of the above cells/structures are **haploid ($n$)**?

(A) Three

(B) Four

(C) Five

(D) Six

**Correct Answer: (C)**

**Worked Solution:** Let us determine the ploidy of each of the eight structures:
1. Protonemal cell of a moss = **Haploid ($n$)**
2. Primary endosperm nucleus in a dicot = Triploid ($3n$)
3. Leaf cell of a moss = **Haploid ($n$)**
4. Prothallus cell of a fern = **Haploid ($n$)**
5. Gemma cell in *Marchantia* = **Haploid ($n$)**
6. Meristem cell of a monocot = Diploid ($2n$)
7. Ovum of a liverwort = **Haploid ($n$)**
8. Zygote of a fern = Diploid ($2n$)
Therefore, exactly **five** structures ((i), (iii), (iv), (v), and (vii)) are haploid ($n$).

**Practice Question 12** Statement-Based MCQ

Read the following statements regarding the female reproductive structures in gymnosperms:

**Statement I:** The megaspore mother cell is differentiated from one of the cells of the nucellus and divides meiotically to form four haploid megaspores.
**Statement II:** One functional megaspore enclosed within the megasporangium develops into a multicellular female gametophyte that bears two or more archegonia.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

**Correct Answer: (A)**

**Worked Solution:** In gymnosperms, the nucellus is protected by envelopes and the composite structure is called an ovule (megasporangium). A megaspore mother cell differentiates from one of the cells of the nucellus and divides meiotically to form four megaspores. One of the megaspores enclosed within the megasporangium develops into a multicellular female gametophyte that bears two or more archegonia (female sex organs). Both statements are completely accurate.

## 11. Common Pitfalls & Rapid Revision Checklist

**High-Alert NEET UG Traps to Avoid:**

- **Trap 1 — Cyanobacteria vs. Algae:** Never classify cyanobacteria ("blue-green algae") under Kingdom Plantae or Algae; they belong to Kingdom Monera.
- **Trap 2 — Isogamy in *Spirogyra* vs. *Ulothrix*:** Both are isogamous, *but Ulothrix* gametes are **flagellated (motile)** whereas *Spirogyra* gametes are **non-flagellated (non-motile)**.
- **Trap 3 — Pyrenoid Composition:** Students often assume pyrenoids are pure starch. Remember: *"Pyrenoids contain protein besides starch."*
- **Trap 4 — Algin vs. Carrageen:** Both are hydrocolloids, but **algin** is produced by **brown algae**, whereas **carrageen** (and agar) is produced by **red algae**.
- **Trap 5 — Rhizoids in Liverworts vs. Mosses:** Liverworts have **unicellular** rhizoids; Mosses have **multicellular and branched** rhizoids.
- **Trap 6 — Origin of Moss Leafy Stage:** The protonema develops directly from a spore, whereas the leafy stage develops from the **secondary protonema as a lateral bud**.
- **Trap 7 — Strobili vs. Heterospory in Pteridophytes:** *Equisetum* forms strobili (cones) just like *Selaginella*, *but Equisetum* is **homosporous** while *Selaginella* and *Salvinia* are **heterosporous**.
- **Trap 8 — *Pinus* vs. *Cycas* Sexuality:** *Pinus* is **monoecious** (male and female cones on the same tree) with branched stems and mycorrhizal roots; *Cycas* is **dioecious** (male cones and megasporophylls on different trees) with unbranched stems and coralloid roots.

**Concise Final Recap**

- **Algae:** Simple, thalloid, autotrophic, largely aquatic; classified into *Chlorophyceae* (chl *a, b*; starch; 2–8 equal apical flagella), *Phaeophyceae* (chl *a, c*, fucoxanthin; laminarin, mannitol; algin wall; 2 unequal lateral flagella), and *Rhodophyceae* (chl *a, d*, r-phycoerythrin; floridean starch; non-motile stages).

- **Bryophytes:** Amphibians of the plant kingdom; haploid gametophyte ($n$) is dominant; diploid sporophyte ( $2n$) (foot, seta, capsule) is attached and dependent on the gametophyte; divided into *Liverworts* (*Marchantia* — gemmae) and *Mosses* (*Funaria, Polytrichum, Sphagnum* — protonema + leafy stage).
- **Pteridophytes:** First vascular land plants (xylem and phloem); diploid sporophyte ($2n$) with true root, stem, and leaves is dominant; gametophyte is a free-living thalloid *prothallus*; heterospory in *Selaginella* and *Salvinia* is the precursor to the seed habit; four classes: *Psilopsida, Lycopsida, Sphenopsida, Pteropsida*.
- **Gymnosperms:** Naked-seeded plants (ovules not enclosed by an ovary wall); heterosporous; highly reduced male (pollen grain) and female gametophytes are *not free-living* and remain retained within sporangia on the sporophyte; wind-pollinated; pollen tube delivers male gametes to archegonia.
- **Angiosperms:** Flowering plants (from microscopic *Wolffia* to tall *Eucalyptus* > $100\ \mathrm{m}$); ovules develop inside flowers and seeds are enclosed in fruits; divided into *Dicotyledons* and *Monocotyledons*.